

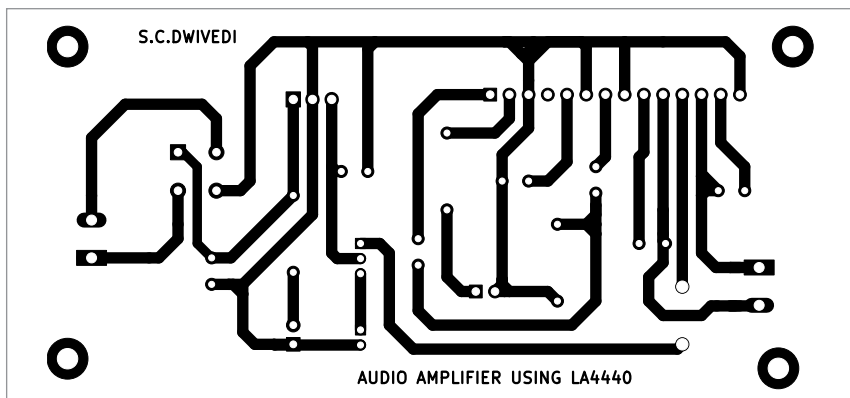


Fig. 3: Actual-size PCB layout

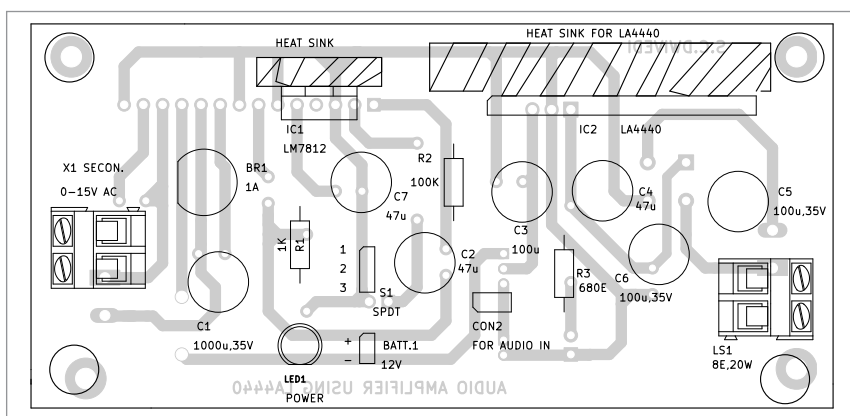


Fig. 4: Component layout of the PCB

Working of the circuit is simple. After connecting 230V AC across CON1 and flipping switch S1 towards AC mains, your circuit is ready to use. But before doing that you have to give audio input across CON2 via desktop/laptop, etc. As soon as you connect audio input and power on, the output would be heard loudly across loudspeaker LS1.

To use the 12V battery as power supply, connect it and flip switch S1

towards battery side. As before, as soon as you connect audio input, the loud output can be heard across loudspeaker LS1.

An actual-size, single-side PCB layout for the audio amplifier circuit is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Fix LED1 and CON2 in front of box and loudspeaker LS1 on its rear side. CON 2 is for audio input

PARTS LIST

Semiconductors:

- IC1 - LM7812, 12V voltage regulator
- IC2 - LA4440 audio amplifier
- BR1 - 1A bridge rectifier
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2 - 100-kilo-ohm
- R3 - 680-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2, C4, C7 - 47 μ F, 35V electrolytic
- C3, C5, C6 - 100 μ F, 35V electrolytic

Miscellaneous:

- LS1 - 8-ohm, 20W speaker
- S1 - SPDT switch
- CON1 - 2-pin connector
- CON2 - Mono jack
- X1 - 230V AC primary to 15V, 1A secondary transformer
- Heatsink for LA4440
- Heatsink for LM7812

that may be taken from desktop, laptop, mobile phone, etc.

Use of heatsinks is recommended for the ICs LA4440 and LM7812. Fix switch S1 on the front side for selecting mains or battery. Connect the 230V AC primary of transformer X1 and 15V AC at the rear side of box. Keep transformer and battery inside the box.

When LED1 is glowing, it means the circuit is working on mains and when it is off, it means the circuit is working on battery. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-audio-amplifier-using-la4440>

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